
Observability, Memory, and the Dynamics of Cooperation in Heterogeneous Multi-Agent Reinforcement Learning

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Abstract

We ask how three design choices—what an agent *observes*, how much history it *remembers*, and which *learning rule* it uses—shape whether reciprocal cooperation emerges in multi-agent reinforcement learning. Using collective reinforcement-learning dynamics (CRLD), the deterministic limit of temporal-difference learning, we read the learning process as a *flow* in policy space and study a heterogeneous Iterated Prisoner’s Dilemma in which one agent observes the full history and the other is selectively blind along controlled dimensions. Three findings emerge. First, *self-observability is the load-bearing perceptual channel*: an agent that cannot see its own last action cannot implement a reciprocal rule and the flow collapses to mutual defection, whereas losing sight of the *partner*—or seeing it only conditionally—is far less damaging. Second, *the cooperative basin is created by memory*: absent at memory one, a cooperative attractor appears at memory two and enlarges as agents condition on more history, a purely deterministic counterpart to the intrinsic-fluctuation route to cooperation. Third, *the learning rule reshapes the attractor*: policy-gradient learning is bistable between full cooperation and full defection, whereas value-based learning settles on a robust interior compromise that avoids total defection—the same game yields different cooperative outcomes under the two rules. Throughout, the dominant failure mode of observational heterogeneity is *shared* defection rather than one agent dominating another. Reading the dynamics as a flow makes each of these mechanisms—and the basins behind them—explicit.

1 Introduction

Cooperation underlies much of human flourishing Hare and Woods [2020], Bar-Yam [2000], yet the complexity of modern societies routinely produces social dilemmas in which individual incentives and collective welfare diverge—from climate policy Kersting [2017] and pandemic response Glennerster et al. [2024] to political stalemate Frieden and Lake [2015], Dafoe et al. [2020] and economic distrust Acemoglu and Robinson [2012]. Such problems are increasingly modeled with Markov Decision Processes and reinforcement learning (RL), which capture how agents—governments, firms, consumers—adapt strategies from partial observations of an evolving environment Glennerster et al. [2024], Charpentier et al. [2021].

A defining feature of real-world actors is that they do *not* perceive their shared environment identically: a firm and a regulator see different slices of the same market, and treaty negotiators hold asymmetric

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information about each other—and about their own commitments. We therefore ask a focused question: *how does heterogeneity in observability shape whether learning agents come to cooperate?* We isolate observability from other heterogeneities (learning rate, risk tolerance) so its effect can be measured cleanly, in the general-sum Iterated Prisoner’s Dilemma Barfuss [2022].

Our approach is to treat the learning process itself as a dynamical system. In the deterministic CRLD limit, the learning rule is a vector field (a *flow*) on the space of joint policies; cooperation corresponds to a basin of attraction, and we can ask directly which design choices create or destroy that basin. This lets us move past a single cooperation statistic to the *geometry* that produces it, and to sweep observability, memory, and learning rule on the same footing. A preliminary report by the present authors over-read these simulations as showing that knowledge-asymmetric agents cooperate readily; the controlled, flow-based analysis here gives a sharper and more mechanistic account.

2 Methods

2.1 Learning dynamics

We simulate two agents, $i \in \{1, 2\}$, each choosing an action $a^i \in \{c, d\}$. Rather than noisy, sample-based simulations, we use *collective reinforcement learning dynamics* (CRLD) Barfuss [2023, 2022], Barfuss et al. [2019], the deterministic, infinite-batch (strategy-averaged) limit of temporal-difference learning: it replaces stochastic value updates with deterministic equations in *strategy space* that are efficient and analytically transparent. Each agent holds a policy $x^i(a \mid o)$ over actions given an *observation* o , which need not coincide with the true state s —the lever for perceptual heterogeneity (Section 2.3). The policy updates along the strategy-averaged temporal-difference error δ^i via the policy-space replicator–mutator map Barfuss [2022], Barfuss et al. [2019] (not a tabular value update):

$$x_{t+1}^i(a \mid o) = \frac{x_t^i(a \mid o) \exp(\alpha_i \beta_i \delta_t^i(o, a))}{\sum_b x_t^i(b \mid o) \exp(\alpha_i \beta_i \delta_t^i(o, b))}, \quad (1)$$

where $\alpha_i \in (0, 1]$ is the learning rate and $\beta_i > 0$ is the choice intensity (inverse temperature) controlling the exploration–exploitation trade-off. Equation (1) replaces the explicit ϵ -greedy action selection and tabular SARSA value update of standard RL; in the deterministic limit, softmax action selection with intensity β_i is folded directly into the policy and its update.

The strategy-averaged temporal-difference error for actor–critic learning is

$$\delta_t^i(o, a) = (1 - \gamma_i) \bar{R}^i(o, a) + \gamma_i \bar{V}_{\text{next}}^i(o, a) - \bar{V}^i(o), \quad (2)$$

where $\gamma_i \in [0, 1)$ is the discount factor, $\bar{R}^i(o, a)$ the strategy-averaged immediate reward, $\bar{V}^i(o)$ the value of observation o under the joint policy, and $\bar{V}_{\text{next}}^i(o, a)$ the strategy-averaged successor value; the $(1 - \gamma_i)$ prefactor puts immediate and future returns on the same scale Barfuss et al. [2019]. Using the expected successor value makes this the deterministic, on-policy form of TD learning (the infinite-batch limit of Expected SARSA Sutton and Barto [2018]); the values $\bar{V}^i$ solve the corresponding Bellman system in closed form. Under partial observability all averages additionally integrate the belief over true states induced by the observation model, following the partially observable CRLD formulation Barfuss and Mann [2022].

The dynamics as a flow. Equations (1)–(2) define a deterministic map on the joint policy—equivalently, a vector field (a *flow*) on strategy space. From any joint policy the learning rule points in a definite direction, the fixed points where the flow vanishes are the learned equilibria, and the *cooperation rate* we report is precisely the measure of the basin of attraction of the cooperative fixed point under random initial policies. Visualizing this flow (Figures 1–2) exposes *why* a scenario does or does not cooperate, complementing the reward statistics with the geometry behind them.

Because α_i , β_i , and γ_i can be set per agent, this framework natively supports *heterogeneous* agents that differ in learning speed and effective risk tolerance. In this paper we hold these parameters equal across agents (Appendix) so that the sole heterogeneity under study is observability.

2.2 Environment

The environment is the Iterated Prisoner’s Dilemma (IPD). The one-shot payoffs (Table 1) are $R = 1$ (mutual cooperation), temptation $T = 1.2$, sucker $S = -0.5$, and punishment $P = 0$, satisfying $T > R > P > S$ and $2R > T + S$. We embed the stage game into a *memory-two* history (an agent’s state is the previous two rounds’ outcomes)—the smallest history rich enough to host the reciprocal strategies (win-stay–lose-shift, tit-for-tat) that direct reciprocity relies on Schmid et al. [2021], Hilbe et al. [2017], yet low-dimensional enough for exhaustive analysis.

Table 1: Payoff matrix for the Iterated Prisoner’s Dilemma (row agent, column agent).

	c	d
c	(1, 1)	(-0.5, 1.2)
d	(1.2, -0.5)	(0, 0)

2.3 Observational heterogeneity

Agent 1 is a *full observer* (perceives both agents’ last two actions without aliasing). Agent 2 is a *partial observer*: along controlled dimensions, parts of the most recent round are masked, aliasing several true states to one observation (its belief is spread uniformly over them) while the prior round stays fully observed. We study four restrictions of Agent 2; exact state-to-observation maps are in the Appendix (Tables 2–5).

- **Self-aware:** Agent 2 remembers its *own* most recent action but not Agent 1’s (it cannot see what the other did last round).
- **Non-self-aware:** Agent 2 remembers Agent 1’s most recent action but not its *own* (it cannot see what it itself did last round).
- **Cooperation-tracking:** Agent 2 perceives Agent 1’s last action only when Agent 1 *cooperated*; Agent 1’s defections are masked.
- **Defection-tracking:** Agent 2 perceives Agent 1’s last action only when Agent 1 *defected*; Agent 1’s cooperations are masked.

We probe each setting two ways: by the *flow geometry* in policy space (Figure 1), and by integrating the deterministic dynamics from $N = 1,200$ random softmax initial policies to a fixed point (tolerance 10^{-5} , $\leq 4,000$ steps). The *cooperation rate* is the fraction of runs reaching mutual cooperation (both cooperation probabilities > 0.9)—the measure of the cooperative basin under random initialization.

3 Results

Self-observability is the load-bearing perceptual channel. Reading the heterogeneous IPD as a flow (Figure 1) shows cooperation living in a basin that observability controls. Under full observability a cooperative fixed point at (1, 1) is reachable, but it shares the space with a large defection basin—about 14% of random initializations cooperate—so even symmetric, complete information does not make cooperation typical. Degrading the partial observer shrinks the cooperative basin, and *which* coordinate is hidden matters far more than how much is hidden. When Agent 2 cannot see its *own* last action (non-self-aware), the flow drains uniformly to defection: an agent that cannot identify its own move cannot “stay” or “shift” relative to it, so the substrate for direct reciprocity is gone. Losing sight of the *partner* instead (self-aware) is far milder—a cooperative basin survives—and merely *conditioning* which of the partner’s moves are visible (cooperation- vs. defection-tracking) is close to full observability. It is self-observability, not partner-observability, that reciprocity is built on. Across initializations the two agents converge to nearly the same policy: the failure mode of heterogeneity is *shared* defection, not one agent exploiting the other (Appendix, Figure 4).

Memory creates and enlarges the cooperative basin. That cooperative basin is not intrinsic to the game—it is built by *memory*. Re-running the full-observability dynamics at memory one, two, and four (Figure 2) shows that at memory one the flow does not reach mutual cooperation from

Heterogeneous observability — learning flow in the Iterated Prisoner's Dilemma (memory-2, actor-critic)

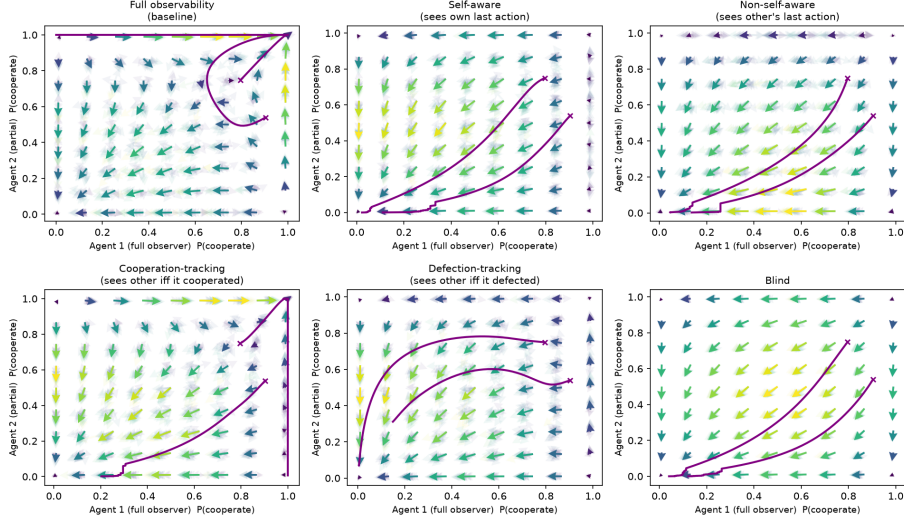


Figure 1: Learning flow of the heterogeneous memory-two IPD (actor-critic CRLD), one panel per observability condition of the partial observer (Agent 2; Agent 1 is always a full observer). Arrows are the deterministic learning update at each joint policy; purple curves are two trajectories; axes are each agent’s cooperation probability. Mutual cooperation is the corner $(1, 1)$, defection $(0, 0)$. A basin draining to cooperation exists under full observability, cooperation-tracking, and—weakly—self-awareness, but the flow collapses toward defection when the partial observer cannot see its *own* last action (non-self-aware) or is blind.

Full observability: the cooperative basin emerges and grows with memory (IPD, actor-critic)

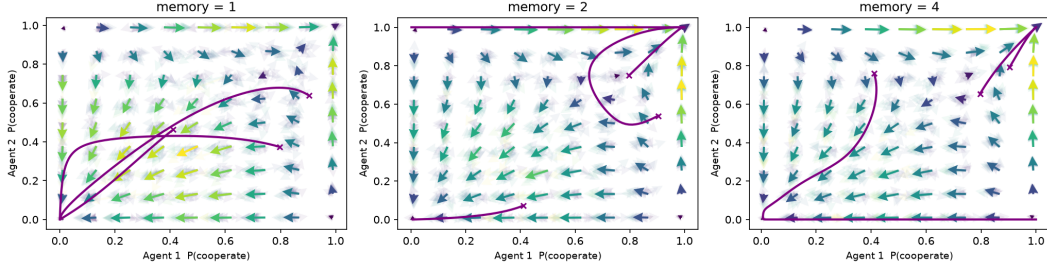


Figure 2: Full-observability IPD learning flow at memory one, two, and four (actor-critic). A cooperative fixed point at $(1, 1)$ is not reached at memory one but becomes an attractor from memory two onward, and its basin enlarges with memory—a deterministic analogue of the intrinsic-fluctuation mechanism Barfuss and Meylahn [2023].

our initializations; a cooperative fixed point at $(1, 1)$ becomes reachable only at memory two, and its basin enlarges at memory four. Longer histories let agents represent richer reciprocal strategies (memory- k tit-for-tat / win-stay-lose-shift) that turn cooperation into a deterministic attractor. This is the deterministic counterpart of the finite-batch *intrinsic fluctuations* known to promote cooperation in the memory-one IPD Barfuss and Meylahn [2023]: memory and sampling noise enlarge the same cooperative basin by different routes.

The learning rule co-determines the attractor. Whether cooperation is reachable at all also depends on the learning rule (Appendix, Figure 3). Policy-gradient (actor-critic) learning is *bistable*: depending on initialization it converges either to mutual cooperation $(1, 1)$ or to mutual defection $(0, 0)$, with most random starts in the defection basin. Value-based (SARSA) learning instead converges, from almost any start, to a single *interior* partial-cooperation fixed point—it never

collapses to total defection, but never reaches full cooperation either. The same game, payoffs, and observability thus produce qualitatively different cooperative outcomes under the two rules, so the choice of algorithm is part of the cooperation question, not an implementation detail. (In anti-coordination variants the two rules select different equilibria outright; see the supplementary flow-grid matrix.)

4 Discussion

Reading multi-agent learning as a flow turns “does cooperation emerge?” into the sharper question of *what shapes the cooperative basin*, and our three results answer it with three distinct knobs. *Observability* sets whether reciprocity is expressible at all: self-observation is the load-bearing channel, because a reciprocal rule is defined relative to one’s own last move, while partner-observation is comparatively dispensable. *Memory* sets whether a cooperative attractor exists: it is absent at memory one and built up as agents condition on more history. *The learning rule* sets the shape of the attractor: policy-gradient is bistable between full cooperation and full defection, whereas value-based learning settles on a robust interior compromise.

A unifying reading is that cooperation here is gated by how much reciprocal structure the learners can both *represent* and *stabilize*. Memory and finite-batch intrinsic fluctuations Barfuss and Meylahn [2023] expand the same basin by different routes—more representable history versus noise that escapes the defection trap—which suggests they are partly interchangeable; whether fluctuations can rescue the non-self-aware case that memory alone cannot is a natural test. Equally, the dominant failure mode throughout is *shared* defection rather than exploitation: perceptual asymmetry costs both agents the cooperative surplus instead of letting the better-informed one dominate.

5 Future work and limitations

Natural next steps are to (i) sweep β and γ to map where, if anywhere, observational heterogeneity becomes cooperation-promoting; (ii) reintroduce finite-batch fluctuations Barfuss and Meylahn [2023] to test whether they rescue cooperation under partial observability where memory alone cannot; and (iii) extend beyond the IPD to public-goods, bargaining, and common-pool games, eventually connecting these low-dimensional dynamics to deep-RL and language-model agents. The present conclusions are correspondingly bounded: the IPD captures a narrow slice of strategic behavior; CRLD assumes deterministic, infinite-batch update rules rather than human idiosyncrasy; outcomes depend on the chosen parameters and on the single regime we report; and cooperation rates are statements about the measure of a basin of attraction under random initial policies, not about any single trajectory.

6 Acknowledgments

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A Appendix / supplemental material

A.1 Software and reproducibility

All experiments use an extension of the open-source pyCRLD library Barfuss [2023] that adds heterogeneous per-agent observation tensors. The deterministic dynamics, observation-tensor construction, and figure-generation scripts used for every result and figure in this paper are released with the camera-ready version.

Relation to an earlier preliminary report. An earlier exploratory writeup by the authors reported that knowledge-asymmetric agents cooperate readily and that better-informed agents earn more. Those conclusions do not survive a controlled analysis: the optimistic cooperation reading came partly from a run with an out-of-range discount factor ($\gamma = 10$, which makes the $(1 - \gamma)$ prefactor negative and the value solve ill-posed) and from per-scenario reporting drift, and the apparent reward advantage of the better-informed agent vanishes—indeed reverses—once a valid $\gamma = 0.9$ is used. All results here use one consistent, flow-based pipeline at $\gamma = 0.9$.

A.2 Parameters

For all experiments we use learning rate $\alpha = 0.1$, discount factor $\gamma = 0.9$, and choice intensity $\beta = 1$, identical across both agents. Memory length is two ($h = (2, 2, 2)$ in the history-embedding convention). Each scenario uses $N = 1,200$ random softmax initial policies; trajectories run up to 4,000 steps with fixed-point tolerance 10^{-5} .

A.3 Learning rule and aggregate statistics

Learning rule and cooperation: full-observability IPD flow (memory-2)

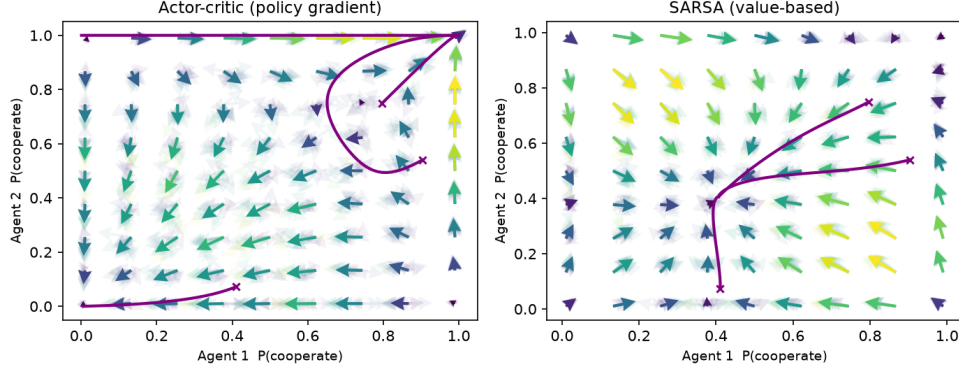


Figure 3: The learning rule reshapes the attractor (full-observability IPD, memory-two). *Left*: policy-gradient (actor-critic) is bistable—initial conditions decide between mutual cooperation $(1, 1)$ and mutual defection $(0, 0)$. *Right*: value-based (SARSA) learning converges from almost any start to a single interior partial-cooperation fixed point, avoiding total defection but never reaching full cooperation.

A.4 Representative trajectory

A.5 Observation maps

Tables 2–5 give the state-to-observation maps for Agent 2 in each scenario. Each row is one of the 16 memory-two states; entries show what Agent 2 perceives, with $*$ denoting a masked (unobserved) coordinate that aliases the state with others sharing the same observation. Agent 1 observes every state without aliasing in all scenarios.

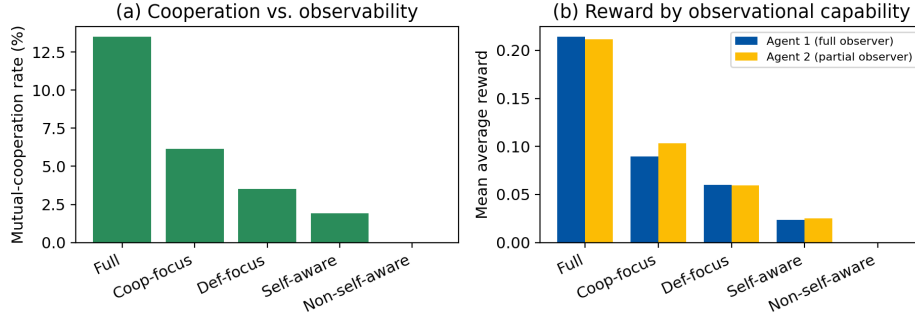


Figure 4: Aggregate over observability scenarios, $N = 1,200$ random initial conditions each. (a) Mutual-cooperation rate decreases monotonically as Agent 2’s observability is degraded, from full observability to the non-self-aware case. (b) Mean reward per agent; the full and partial observers earn nearly the same reward—the gap is small and both fall far short of the cooperative payoff $R = 1$ (shared defection, not exploitation).

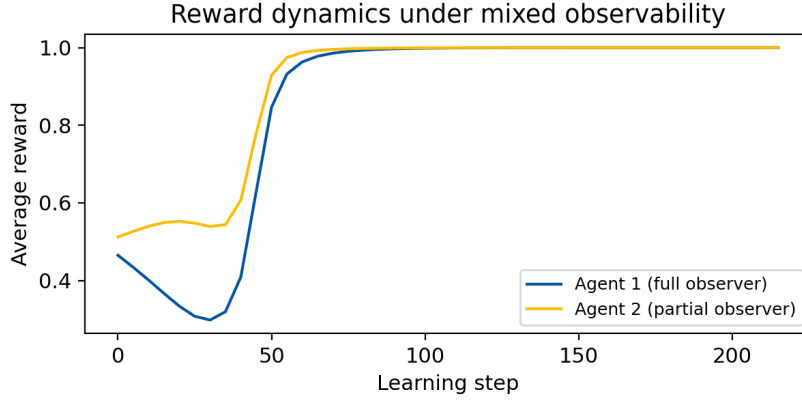


Figure 5: A representative learning trajectory under mixed observability. Average reward per agent versus learning step; the agents settle near the defection payoff, consistent with the aggregate finding that defection is the modal attractor.

Table 2: Self-aware Agent 2: keeps its own most recent action, masks Agent 1’s.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(*, C)
(C, C)	(C, D)	(C, C)	(*, D)
(C, D)	(C, C)	(C, D)	(*, C)
(C, D)	(C, D)	(C, D)	(*, D)
(D, C)	(D, C)	(D, C)	(*, C)
(D, C)	(D, D)	(D, C)	(*, D)
(D, D)	(D, C)	(D, D)	(*, C)
(D, D)	(D, D)	(D, D)	(*, D)

Table 3: Non-self-aware Agent 2: keeps Agent 1's most recent action, masks its own.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, *)
(C, C)	(C, D)	(C, C)	(C, *)
(C, D)	(C, C)	(C, D)	(C, *)
(C, D)	(C, D)	(C, D)	(C, *)
(D, C)	(D, C)	(D, C)	(D, *)
(D, C)	(D, D)	(D, C)	(D, *)
(D, D)	(D, C)	(D, D)	(D, *)
(D, D)	(D, D)	(D, D)	(D, *)

Table 4: Cooperation-tracking Agent 2: sees Agent 1's last action only when Agent 1 cooperated.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, C)
(C, C)	(C, D)	(C, C)	(C, D)
(C, C)	(D, C)	(C, C)	(*, C)
(C, C)	(D, D)	(C, C)	(*, D)
(D, C)	(C, C)	(D, C)	(C, C)
(D, C)	(C, D)	(D, C)	(C, D)
(D, C)	(D, C)	(D, C)	(*, C)
(D, C)	(D, D)	(D, C)	(*, D)

Table 5: Defection-tracking Agent 2: sees Agent 1's last action only when Agent 1 defected.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(*, C)
(C, C)	(C, D)	(C, C)	(*, D)
(C, C)	(D, C)	(C, C)	(D, C)
(C, C)	(D, D)	(C, C)	(D, D)
(D, C)	(C, C)	(D, C)	(*, C)
(D, C)	(C, D)	(D, C)	(*, D)
(D, C)	(D, C)	(D, C)	(D, C)
(D, C)	(D, D)	(D, C)	(D, D)